SURROGATE-ASSISTED pCI-GRADIENT ASCENT

A Combinatorial Optimization Procedure for Reforming Knowledge Organization Systems

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In my research document, *From Literary Warrant to Cognitive Warrant*, I introduced **structural cognitive information** (in Section 3.5) as the total uncertainty reduction achieved by the full structural signature of the heading system—heading congruence and hierarchical depth jointly, with the following decomposition:

$$\begin{aligned}
\mathrm{CI}_{\text{struct}}(d) \;\equiv\; I(Y;\, S, h \mid d) &= H(Y \mid d) - H(Y \mid d, S, h) \\
&= H(Y \mid d) - \big[H(Y \mid d, S) - I(Y;\, h \mid d, S)\big] \\
&= \big[H(Y \mid d) - H(Y \mid d, S)\big] + I(Y;\, h \mid d, S) \\
&= \quad \underbrace{I(Y;\, S \mid d)}_{\text{CI}(d)\text{: boundary increment}} \quad + \underbrace{I(Y;\, h \mid d,\, S)}_{\text{hierarchical increment}}
\end{aligned}$$

The **hierarchical increment** $I(Y; h \mid d, S)$ is the additional uncertainty reduction from knowing how far into the heading structure a boundary crossing reaches, beyond what is provided by knowing the continuous distance and whether any boundary was crossed at all. It measures the cognitive value of polyhierarchical organization as a quantity over and above gross categorical partitioning.

In the same section, I presented the **pointwise cognitive information** metric (pCI). For a specific stimulus pair with observed values (d_i, S_i, Y_i), pCI_i is defined as:

$$\mathrm{pCI}_i = \log \frac{P(Y_i \mid d_i,\, S_i)}{P(Y_i \mid d_i)}$$

This is the pointwise mutual information between the heading congruence and the sense identity for item i (a given pair of prime and target sentences), conditional on the observed distance. It quantifies how much the specific LCSH heading assignment for this pair shifted the posterior probability toward the correct sense identification, relative to the baseline established by continuous similarity alone. Subject headings with a negative pCI are inconsistent with the behavioral response — the LCSH categorization shifted posterior probability away from the observed outcome.

For this document, we derive a new metric. The derivation aligns with the derivation of pCI from CI(d). Thus, from structural cognitive information, we have, at the item level, its pointwise derivative:

$$\mathrm{pCI}_i^{\mathrm{struct}} \;=\; \log \frac{P(Y_i \mid d_i,\, S_i,\, h_i)}{P(Y_i \mid d_i)}.$$

Although the pCI and pCI$^{\mathrm{struct}}$ diagnostics identify *which* LCSH heading assignments are cognitively problematic; they do not, by themselves, specify *what* a cataloger or editor of a controlled vocabulary should do. A heading with persistently negative pCI is known to be actively misleading, but the appropriate corrective action—reassigning the sense to a different authority record, splitting one record into two, merging neighboring records, or restructuring BT/NT links—is not uniquely determined by the diagnostic score alone. This portion of my theoretical framework develops a formal procedure for selecting among these actions by extending the cognitive information framework into a combinatorial optimization setting.

1. Graph Surgery Under an Information-Theoretic Objective

Let $G = (V, E)$ denote the *directed acyclic graph* (DAG) representing the full LCSH BT/NT polyhierarchy, where each vertex $v_i \in V$ corresponds to an authority record and each directed edge $uv \in E$ encodes a broader-term link ($u \to v$ means v is a BT of u, i.e., u is narrower than v). G is a *simple graph*: each ordered pair of vertices appears at most once as an edge.

Given a problematic vertex v_0 identified by negative pCI, we define its **reform neighborhood** $N(v_0, D)$ as the *induced subgraph* of G containing all vertices within hierarchical distance $D = 3$ of v_0:

$$V(N) = \big\{ v_i \in V(G) \;\big|\; \texttt{hier\_distance}(v_0,\, v_i) \le D \big\},$$
$$E(N) = \big\{ uv \in E(G) \;\big|\; u, v \in V(N) \big\}.$$

The restriction to $D = 3$ is consistent with the threshold for `same_nbhd` and ensures computational tractability; N is finite and typically contains on the order of tens to low hundreds of vertices for any LCSH topical heading. Vertices and edges outside N are held fixed during the optimization.

The **reform objective** is to maximize the total pointwise cognitive information[1] across the neighborhood:

$$\mathcal{F}(N) \;=\; \sum_{v_i \in V(N)} \mathrm{pCI}_{v_i} \;=\; \sum_{v_i \in V(N)} \log \frac{P(Y_i \mid d_i,\, S_i,\, h_i)}{P(Y_i \mid d_i)}.$$

Equivalently, the objective can be stated at the system level in terms of expected cognitive information $\overline{\mathrm{CI}}$, which is the weighted aggregate of pCI values over the empirical distance distribution and therefore increases whenever the neighborhood-level sum increases. The two formulations are interchangeable for the local search procedure developed below.

Allowable edit operations on N are:

[1]Throughout this document, pCI denotes the structural extension $\mathrm{pCI}^{\mathrm{struct}}$. This extension is required here because graph edits act on both S_i and h_i; a surrogate restricted to S_i alone would be blind to edits (e.g., edge reassignment) that change hierarchical proximity without changing heading congruence. The unmarked notation pCI is retained throughout for readability and should be read as $\mathrm{pCI}^{\mathrm{struct}}$.

1. **Add or remove a vertex** (creating a new authority record or consolidating an existing one into a neighbor).
2. **Add, remove, or redirect an edge** (establishing, eliminating, or changing the endpoint of a BT/NT link).

Common compound operations such as node splitting or node merging can be decomposed into a sequence of these "atomic" add/remove steps.

2. The Surrogate: The Fitted Mixed Model as a Predictive Instrument

The central obstacle to optimizing $\mathcal{F}(N)$ is that pCI_{v_i} is defined in terms of behavioral probabilities $P(Y_i \mid d_i, S_i, h_i)$ and $P(Y_i \mid d_i)$, both of which are estimated from experimental data. It is computationally infeasible—and practically impossible—to re-run a behavioral experiment after each proposed graph edit. A **surrogate model** is therefore required: a fitted function $\hat{f}$ that predicts the behavioral probability for any configuration of predictor values without new data collection.

The key structural insight is that graph edits affect pCI exclusively through two predictors: the binary heading congruence indicator S_i (`same_heading`) and the hierarchical distance h_i (`hier_distance`). The continuous semantic predictor `distance_bert` is invariant to graph edits—it is determined by the linguistic contexts of the word senses, not by the LCSH structure. This means that after any edit ε applied to N, the new values S_i^{ε} and h_i^{ε} are deterministic functions of the edited graph structure, computable in $O(|V(N)| + |E(N)|)$ time via standard *shortest-path algorithms.*

The fitted cross-classified mixed-effects model is the surrogate. Let $\phi_i(\cdot)$ denote the linear predictor of the model evaluated at a given configuration of predictors for item i, and let $\sigma(\cdot) = (1 + e^{-(\cdot)})^{-1}$ denote the logistic function. The model predicts the behavioral probability that item i is correctly disambiguated as:

$$\hat{P}(Y_i \mid d_i,\, S_i,\, h_i) \;=\; \sigma(\phi_i(d_i,\, S_i,\, h_i))\,, \tag{1}$$

where ϕ_i includes the full fixed-effect specification—the main effects of `distance_bert`, `same_heading`, and `hier_distance`, together with the nonlinear and interaction terms from the H3 specification—plus the item-level random intercept $\hat{u}_i$. The marginal probability $\hat{P}(Y_i \mid d_i)$, obtained by evaluating Equation 1 with $S_i = 0$ and $h_i = h_i^{\varnothing}$ (the hierarchical distance that would obtain under no categorical boundary), serves as the distributional baseline.

2.1. Operationalizing the Marginal Baseline

The marginal probability $\hat{P}(Y_i \mid d_i)$ requires care in definition. It is not simply the model's fitted value with S_i set to zero, because h_i (`hier_distance`) is itself a function of the graph structure and therefore not independent of S_i. For a pair with $S_i = 0$ (different authority records, no shared heading), h_i is already defined as the BT-hop path length between each of the two assigned headings to their lowest common ancestor in G, and the marginal baseline is obtained by evaluating Equation 1 at the observed $(d_i, S_i = 0, h_i)$ directly. No adjustment is required.

For a pair with $S_i = 1$ (both senses assigned to the same authority record), the marginal baseline requires a counterfactual: what hierarchical distance $h_i^{\varnothing}$ would obtain if the boundary were removed? Since both senses map to the same node in G, the observed $h_i = 0$ by construction; removing the boundary does not change the graph and therefore does not change h_i. The marginal baseline for such pairs is thus evaluated at $(d_i, S_i = 0, h_i = 0)$—the model's predicted probability for a pair at the same semantic distance with no categorical boundary and zero hierarchical separation.

This asymmetry has a natural interpretation. For cross-boundary pairs ($S_i = 0$), the marginal baseline is the model's prediction absent categorical information but retaining the hierarchical signal already present in h_i. For same-heading pairs ($S_i = 1$), the marginal baseline is the model's prediction at maximum hierarchical proximity with no categorical signal—the most conservative possible baseline, against which the heading's positive contribution to disambiguation is measured. The pointwise cognitive information $\widehat{\mathrm{pCI}}_i$ for same-heading pairs is therefore a direct measure of how much the categorical boundary adds over the null condition of zero structural separation, which is precisely the quantity of interest for reform triage: headings with negative pCI in this regime are actively misleading relative to a catalog that provided no categorical signal at all.

The **predicted pCI** for item i under edit ε is then:

$$\widehat{\mathrm{pCI}}_i^{\varepsilon} = \log \frac{\hat{P}(Y_i \mid d_i,\, S_i^{\varepsilon},\, h_i^{\varepsilon})}{\hat{P}(Y_i \mid d_i)}, \tag{2}$$

and the predicted *change* in pCI induced by the edit is:

$$\begin{aligned} \Delta\widehat{\mathrm{pCI}}_i(\varepsilon) &= \widehat{\mathrm{pCI}}_i^{\varepsilon} - \widehat{\mathrm{pCI}}_i^{\mathrm{pre}} \\ &= \log \frac{\hat{P}(Y_i \mid d_i,\, S_i^{\varepsilon},\, h_i^{\varepsilon})}{\hat{P}(Y_i \mid d_i,\, S_i,\, h_i)}. \end{aligned} \tag{3}$$

Both quantities are computable in closed form from the fitted model's `predict()` output applied to the pre- and post-edit predictor configurations; no new experimental data are required. The predicted change in the neighborhood objective function is then:

$$\Delta\hat{\mathcal{F}}(\varepsilon) \;=\; \sum_{v_i \in V(N)} \Delta\widehat{\mathrm{pCI}}_i(\varepsilon). \tag{4}$$

2.2. Relationship to the Linear Approximation

For small, local edits in which $|\Delta S_i^\varepsilon|$ and $|\Delta h_i^\varepsilon|$ are both small relative to the curvature of the logistic function at $\hat{P}(Y_i \mid d_i, S_i, h_i)$, Equation 3 reduces to the first-order approximation:

$$\Delta\widehat{\mathrm{pCI}}_i(\varepsilon) \;\approx\; \hat{\beta}_S \cdot \Delta S_i^\varepsilon \;+\; \hat{\beta}_{\text{hier}} \cdot \Delta h_i^\varepsilon \;+\; \hat{\beta}_{S\times d} \cdot \big(\Delta S_i^\varepsilon \cdot d_i\big). \tag{5}$$

This approximation is adequate when the pre-edit fitted probability $\hat{P}(Y_i \mid d_i, S_i, h_i)$ is well away from the boundaries of the unit interval—the regime in which the logistic function is approximately linear in its argument. It is *not* adequate for the reform scenarios of greatest practical interest. Node-splitting operations that create a new authority record for one sense of a problematic heading set $\Delta S_i^\varepsilon = -1$ for all pairs that previously shared the heading; edge-reassignment operations that move a heading to a hierarchically distant parent can produce $|\Delta h_i^\varepsilon|$ spanning several BT levels. In both cases the discrete jump is large, the logistic nonlinearity is consequential, and Equation 3 is the appropriate expression. The linear approximation Equation 5 is retained here as a closed-form reference for interpretive purposes and for sensitivity analysis under small perturbations, but *Equation 3 is the operationally correct form for the pCI-gradient ascent algorithm.*

3. The Algorithm: pCI-Gradient Ascent on the Reform Neighborhood

With the surrogate in place, the optimization problem reduces to a *combinatorial local search* on the finite space of graphs obtainable from N by allowable single-step edits. The algorithm proceeds as follows.

Step 1. Initialize. Set $N_0 = N$. Compute the current neighborhood objective $\hat{\mathcal{F}}(N_0)$ using item-level fitted values from the baseline mixed model.

Step 2. Enumerate legal moves. At iteration t, construct the set $\mathcal{E}(N_t)$ of all single-step edit operations on N_t that (a) respect LCSH editorial conventions (no isolated vertices, each authority record retains at most one authorized heading label) and (b) preserve the DAG property. The acyclicity constraint is enforced by a depth-first reachability check: an edge addition $u \to v$ is legal only if u is not reachable from v in N_t.

Step 3. Score each move. For each $\varepsilon \in \mathcal{E}(N_t)$, compute $\Delta\hat{\mathcal{F}}(\varepsilon)$ using Equation 3. Propagating effects are handled by re-evaluating S_i^{ε} and h_i^{ε} for *all* pairs in N whose boundary assignments or hierarchical distances are affected by the edit, not only those incident to v_0. This is the $O(|V(N)|^2)$ component of the algorithm; it remains tractable for $D = 3$ neighborhoods.

Step 4. Select the best move. Set $\varepsilon^* = \arg\max_{\varepsilon \in \mathcal{E}(N_t)} \Delta\hat{\mathcal{F}}(\varepsilon)$.

Step 5. Update or terminate. If $\Delta\hat{\mathcal{F}}(\varepsilon^*) > 0$, apply ε^*: set $N_{t+1} = N_t \oplus \varepsilon^*$ and return to Step 2. If $\Delta\hat{\mathcal{F}}(\varepsilon^*) \leq 0$, terminate; the current configuration is a local maximum under the surrogate objective.

This procedure is a *greedy local search* on a discrete graph space with an information-theoretic objective function, using the fitted behavioral model as the evaluation oracle. Convergence to a local maximum is guaranteed because: the objective is bounded above (by the redundancy bound $\mathrm{CI}(d) \leq H(S, h \mid d)$); each accepted move strictly increases $\hat{\mathcal{F}}$; and the move space is finite given the constraint that N remains bounded.

Relation to Standard Optimization

The algorithm is structurally analogous to gradient ascent on a continuous objective, with the discrete move set $\mathcal{E}(N_t)$ playing the role of the gradient direction and $\Delta\hat{\mathcal{F}}$ playing the role of the directional derivative. In the design-of-experiments (DOE) analogy, each iteration is an *one-factor-at-a-time* (OFAT) experiment in which the single factor with the highest predicted main effect is selected and applied—a sequential fractional design over the binary factor space of vertex and edge presence/absence.

4. Edit Type Classification and Interpretive Mapping

The four qualitatively distinct edit types in $\mathcal{E}(N)$ each have a distinct information-theoretic interpretation and a corresponding LCSH-specific editorial action.

Edge reassignment (redirecting the BT link of v_0 to a different parent) increases $\Delta\hat{\mathcal{F}}$ when the current parent is hierarchically distant in BERT-space from the senses mapped to v_0—that is, when the current boundary location is not correlated with semantic structure. This is the most conservative edit: the authority record itself is preserved; only its placement in the hierarchy changes.

Node splitting ($v_0 \to v_0^{(a)}, v_0^{(b)}$) is the optimal compound edit when the senses currently collocated under v_0 are semantically heterogeneous: high variance in `distance_bert` within the heading indicates that the heading is conflating multiple distinct regions of semantic space. The appropriate partition of senses into the two daughter nodes is determined by spectral bisection on the BERT distance matrix restricted to the set of senses currently assigned to v_0—a normalized graph cut that maximizes within-cluster cohesion and between-cluster separation (Shi & Malik, 2000; Urschel & Zikatanov, 2014).

Edge removal (deleting a BT/NT link while retaining both endpoints) increases $H(S \mid d)$, the residual entropy of the boundary given continuous distance, and is appropriate when an existing link is creating spurious near-boundary relationships that inflate S_i for pairs that are semantically distant. This is the move indicated when pCI ≈ 0 and the redundancy bound $H(S \mid d)$ is near-zero: the system is behaving as a threshold on d and the link is informationally inert.

Node merging (collapsing v_0 into a neighbor) is the inverse of splitting and is the appropriate compound edit when two adjacent authority records are semantically redundant—high cosine similarity between their mapped sense embeddings and low $H(S \mid d)$ between them. In LCSH terms, this corresponds to proposing that two authority records be consolidated under a single heading with an expanded scope note.

For *node splitting*, the partition of existing BT links between $v_0^{(a)}$ and $v_0^{(b)}$ is itself a decision variable. The algorithm assigns tentative full inheritance to each as the default and evaluates BT reassignment moves in the immediately following iterations; the default assignment may be overridden based on consultation with the cataloger before continuing. For *node merging*, the merged node v_* initially inherits the union of BT and NT links from both source nodes; conflicting or redundant links are candidates for removal in subsequent iterations, detectable by their negative contribution to $\Delta\hat{\mathcal{F}}$. After the procedure terminates at a local maximum, the authorized labels for split or merged nodes will be the editorial decision of the cataloger.

Table 1 summarizes the four edit types with their information-theoretic signatures, the algorithmic method for identifying the optimal operationalization of each, and the corresponding LCSH editorial action.

Table 1: Classification of edit operations in the pCI-gradient ascent procedure. Each edit type is characterized by the information-theoretic condition that makes it the optimal move, the method for determining its operationalization, and the corresponding LCSH editorial action.

Edit type	**Indicated when**	**Operationalization method**	**LCSH editorial action**
Edge reassignment	Current parent is semantically distant from v_0's mapped senses; ΔS_i is large for high-p$\hat{\text{C}}$I pairs	Enumerate candidate parents by BERT-distance proximity; select by $\arg\max \Delta\hat{\mathcal{F}}$	Revise BT link; update authorized heading label if necessary
Node splitting	High within-heading `distance_bert` variance; multiple sense clusters under v_0	Spectral bisection on within-v_0 BERT distance matrix	Establish two new authority records; assign NT links; adjust scope notes
Edge removal	pCI ≈ 0 and $H(S, h \mid d) \approx 0$; link is structurally redundant	Check $\Delta H(S \mid d)$ after removal; accept if positive and $\Delta\hat{\mathcal{F}} > 0$	Remove BT/NT link; retain both authority records as orphan-adjacent nodes
Node merging	Two adjacent records are semantically redundant; low between-record `distance_bert`	Check cosine similarity of sense embeddings across the two records; accept if above threshold τ	Consolidate into single authority record; expand scope note; reassign NT links

5. Relationship to Existing Methods

The procedure described above is novel as a KOS editing instrument, but it draws on and extends several well-established frameworks in combinatorial optimization and network science. Situating the contribution precisely requires distinguishing what is borrowed from existing work and what is genuinely new.

Surrogate-assisted combinatorial optimization. The two-stage structure of (i) fitting a surrogate to expensive behavioral evaluations and (ii) optimizing over discrete structures using that surrogate is the core methodology of *Bayesian optimization over discrete spaces* (Snoek et al., 2012) and has been applied to molecular graph optimization in drug discovery (Gómez-Bombarelli et al., 2018) and neural architecture search in machine learning (Elsken et al., 2019). The present application is the first, to my knowledge, to apply this two-stage structure to knowledge organization system editing, and the first to use a psycholinguistic mixed-effects model as the surrogate.

Information-theoretic community detection. The global version of the reform problem—finding the partition of all of LCSH that maximizes $\overline{\mathrm{CI}}$ across the full vocabulary—is related to the *map equation* framework of Rosvall & Bergstrom (2008), which minimizes the expected description length of a random walk on a network and thereby identifies the partition that best captures flow structure. Infomap and the present framework share a common information-theoretic ancestry (both derive from the Shannon entropy of a probability distribution over graph paths), but differ in their objective: Infomap optimizes topological flow structure, while the present framework optimizes predicted behavioral disambiguation performance. The local, neighborhood-bounded search developed here is computationally tractable where full-vocabulary Infomap is not.

Ontology repair. The closest domain-specific precursor is the *ontology repair* literature in description logic and the semantic web, specifically the Automated Explanation and Repair of Ontologies (AERO) framework (Schlobach et al., 2007) and its successors. These methods define edit procedures for restoring logical consistency to OWL ontologies given identified inconsistencies—a structurally parallel problem, with logical consistency playing the role of pCI as the objective. The present framework extends this tradition by substituting an empirically grounded, behaviorally validated objective for a purely formal one.

Tree edit distance. For the restricted case in which only edge reassignment is permitted (preserving the set of nodes while changing parent–child relationships), the optimization problem is equivalent to finding the minimum-cost sequence of edit operations transforming one rooted tree into another under a distance metric. The classical Zhang & Shasha (1989) tree edit distance algorithm computes this optimally; the present framework replaces the symmetric distance metric with the asymmetric, directional objective $\Delta\hat{\mathcal{F}}$, which breaks the symmetric assumptions required by the Zhang–Shasha algorithm but is handled naturally by the greedy local search.

6. Local Optima, Metaheuristics, and Practical Scope

Greedy local search on a combinatorial space is guaranteed to reach only a *local* optimum under the surrogate objective, not a global one. Two standard metaheuristics are available to improve solution quality at the cost of additional computation.

Simulated annealing (Kirkpatrick et al., 1983) accepts non-improving moves with probability $e^{-\Delta/T}$, where T is a temperature parameter that decreases over iterations. This allows the search to escape shallow local optima. In the KOS reform context, simulated annealing has a natural interpretation: accepting a temporarily worse configuration corresponds to a cataloger making a speculative edit that degrades local coherence in the short term but opens the path to a more globally coherent structure.

Tabu search (Glover, 1986) maintains a short-term memory of recently visited configurations and prohibits revisiting them, thereby forcing the search to explore new regions of the edit space. In LCSH terms, the tabu list is a record of recently attempted edits that failed to improve $\hat{\mathcal{F}}$—a principled "do not try this again" list for vocabulary editors.

For practical editorial use, the greedy local search is the appropriate starting point: it is interpretable, its output is a single ranked sequence of edit recommendations, and it terminates with a clear stopping criterion. Metaheuristic extensions are more appropriate for research contexts in which the goal is to characterize the space of possible reforms rather than to generate a single actionable recommendation.

7. From Reform Diagnostic to Reform Prescription: Summary

The pCI diagnostic introduced in Section 3.5 of my research document answers the question: *which headings are cognitively problematic?* The pCI-gradient ascent procedure developed here answers the complementary question: *what should be done about them?*

Together, the two instruments constitute a complete empirical reform pipeline for knowledge organization systems:

1. Fit the cross-classified mixed-effects model on behavioral data to obtain $\hat{\beta}_S$, $\hat{\beta}_{\text{hier}}$, and interaction coefficients.
2. Compute item-level $\widehat{\text{pCI}}_i$ for all sense pairs; rank authority records by their mean pCI across associated pairs.
3. For each authority record with mean $\widehat{\text{pCI}} < 0$, define the $D = 3$ reform neighborhood and apply pCI-gradient ascent using the fitted model as surrogate.
4. Output the sequence of edit operations $(\varepsilon_1^*, \varepsilon_2^*, \ldots)$ ranked by $\Delta\hat{\mathcal{F}}$; present to vocabulary editors as an empirically prioritized, action-specific reform agenda.

The pipeline is executable from study output without additional data collection, scales to any controlled vocabulary for which behavioral data and a BT/NT (hierarchical) graph are available, and produces recommendations that are interpretable to catalogers without quantitative training—each recommendation is a specific, named edit to a specific, named authority record, with a predicted improvement in cognitive warrant expressed in units of pCI.

This represents a qualitative advance over the current state of KOS reform practice, in which revision proposals are generated through expert judgment, committee deliberation, and literary warrant analysis. Those processes are not supplanted by the computational procedure: literary warrant, cultural warrant, and professional expertise remain necessary inputs to any final editorial decision. What the procedure adds is a quantitative, behaviorally grounded prior over the space of possible reforms—a principled first filter that directs expert attention toward the changes most likely to improve the cognitive accessibility of the vocabulary for its users.

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